

On Krylov subspace approximations to the matrix exponential operator

Purple team

Marek Fürst, Ján Kovačovský, Ida Nyárfás, Jaroslav Paidar, Petr Rašek

Introduction

- Approximation of $\exp(\tau A)v$ using Krylov methods, A big matrix, τ step size
- Error bounds for Arnoldi and Lanczos algorithms
- Faster convergence than $(I - \tau A)^{-1}v$ for stiff ODE problems, new integration methods
- Bound $\frac{\|\tau A\|^m}{m!}$, superlinear convergence for $m \gg \|\tau A\|$
- Better behaviour for matrix classes: HND, skew-Hermitian, dependent on eigenvalues distribution

Matrix function definition

- $AV_m = V_m H_m + h_{m+1,m} v_m e_m^T, H_m = V_m^* A V_m$
- $q_{m-1}(A)v = V_m q_{m-1}(H_m)e_1 \implies (\lambda I - A)^{-1}v \approx V_m(\lambda I - H_m)^{-1}e_1$
- Numerical range $\mathcal{F}(A) = \{x^* A x : x \in \mathbb{C}^N, \|x\| = 1\}$, $\mathcal{F}(H_m) \subset \mathcal{F}(A)$
- $f(A)v = \frac{1}{i2\pi} \int_{\Gamma} f(\lambda) V_m (\lambda I - H_m)^{-1} e_1 d\lambda$, Γ is contour around $\mathcal{F}(A)$
- $f(A)v \approx V_m f(H_m) e_1$

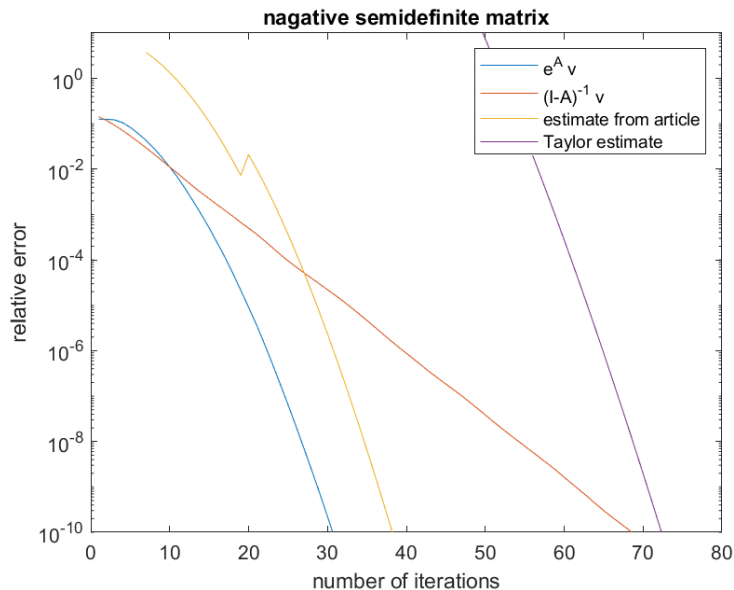
Error bound lemma

- $\|f(A)v - V_m f(H_m) e_1\| \leq \frac{M}{2\pi} \int_{\Gamma} |f(\lambda) - q_{m-1}(\lambda)| |\phi(\lambda)|^{-m} |d\lambda|$
- ϕ conformal mapping from exterior of E to exterior of unit circle
- Γ piecewise smooth, f analytic and continuous to boundary
- $M = l(\partial E)/[d(\partial E)d(\Gamma)]$, $l(\partial E)$ length of boundary, $d(S)$ distance between $\mathcal{F}(A)$ and subset S , E convex closed bounded set in complex plane
- Proof idea
- $\|(\lambda I - A)^{-1} v - V_m(\lambda I - H_m)^{-1} e_1\| \leq c \|p_m(A)\|, p_m(\lambda) = 1$
- $p_m(A) = \frac{1}{2\pi i} \int_{\partial E} p_m(z)(zI - A)^{-1} dz \implies \|p_m(A)\| \leq c \max_{z \in E} |p_m(z)|$
- $\max_{z \in E} |p_m(z)| \leq c |\phi(\lambda)|^{-m}$

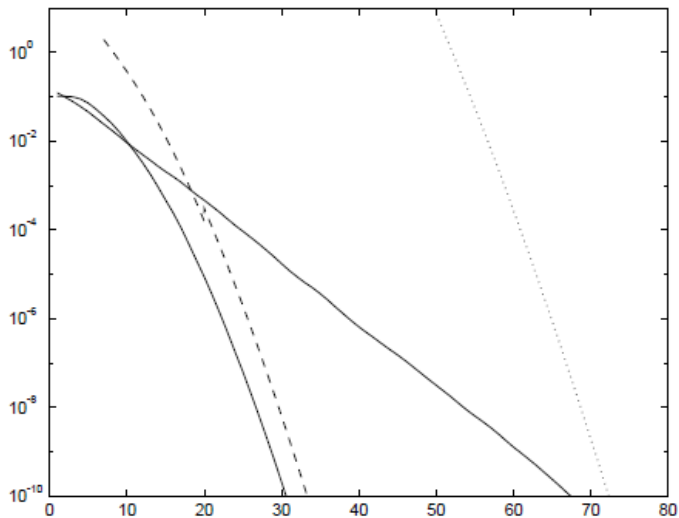
Error bound for Hermitian negative semidefinite matrix

- Eigenvalues in $[-4\rho, 0]$, $\varepsilon_m = \|\exp(\tau A)v - V_m \exp(\tau H_m)e_1\|$
- $\varepsilon_m \leq 10e^{-m^2/(5\rho\tau)}$ for $\sqrt{4\rho\tau} \leq m \leq 2\rho\tau$
- $\varepsilon_m \leq 10(\rho\tau)^{-1}e^{-\rho\tau} \left(\frac{e\rho\tau}{m}\right)^m$ for $m \geq 2\rho\tau$
- Example: A diagonal with eigenvalues in $[-40, 0]$, v random vector of dimension 1001
- Figure: Convergence compared with linear decay for CG $(I - A)^{-1}v$ and theoretical bound (theorem and Taylor remainder)

Error bound for negative semidefinite matrix



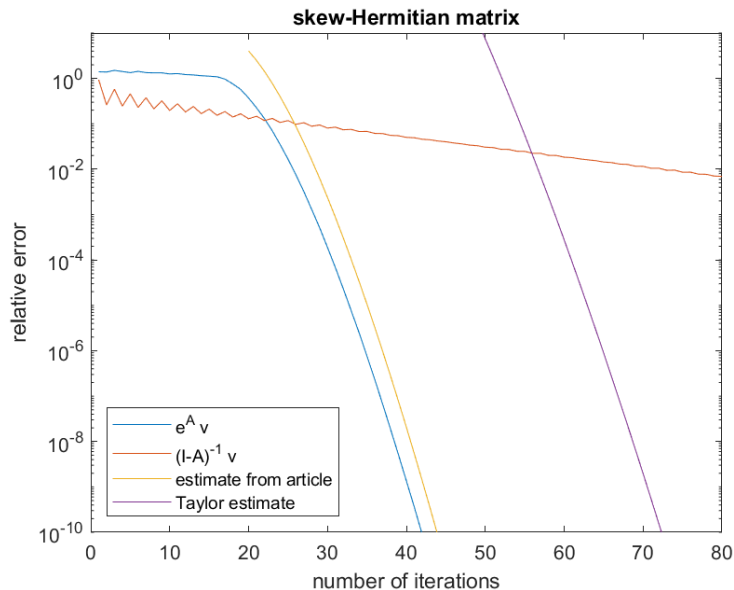
Error bound for negative semidefinite matrix source



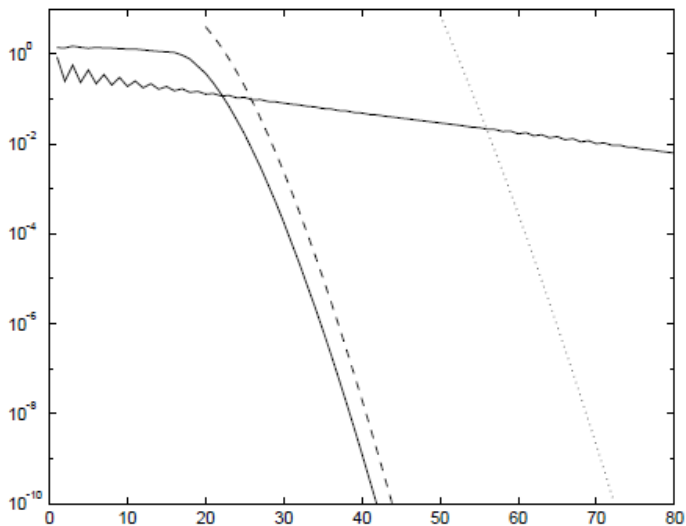
Error bound for skew-Hermitian matrix

- Eigenvalues on imaginary axis in interval of length 4ρ
- $\varepsilon_m \leq 12e^{-(\rho\tau)^2/m} \left(\frac{e\rho\tau}{m}\right)^m$ for $m \geq 2\rho\tau$
- Example: A diagonal with eigenvalues in $[-20i, 20i]$, v random vector of dimension 1001
- Figure: Convergence compared with linear decay for BiCG $(I - A)^{-1}v$ and theoretical bound (theorem and Taylor remainder)

Error bound for skew-Hermitian matrix



Error bound for skew-Hermitian matrix source



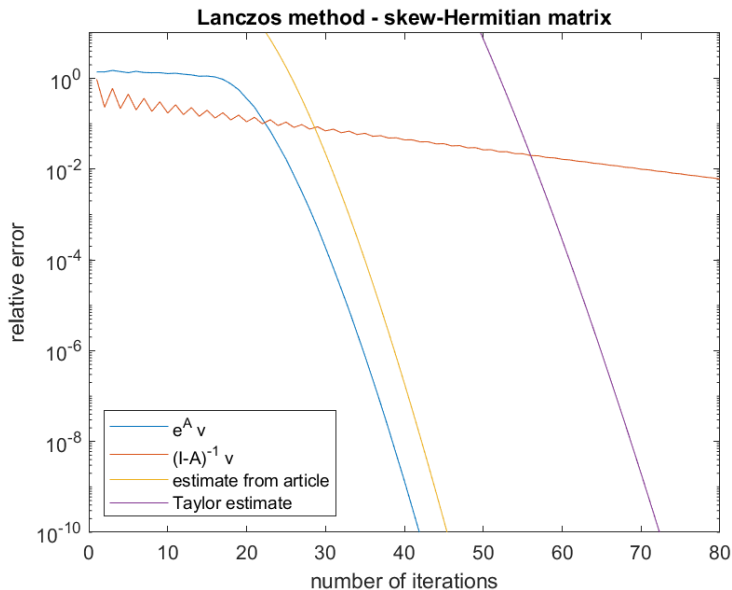
Lanczos method

- Second basis $W_m = (w_1, \dots, w_m)$ of Krylov subspace $K_m(A^*, w_1)$
- Biorthogonality condition $(v_i, w_j) = \delta_{ij}$, $D_m = W_m^* V_m$ block diagonal
- $H_m = D_m^{-1} W_m^* A V_m$ block triangular, short recurrences
- $\mathcal{F}(H_m) \not\subset \mathcal{F}(A)$, oblique projection
- pseudospectrum $\Lambda_\varepsilon(A) = \{\lambda \in \mathbb{C} : \|(\lambda I - A)^{-1}\| \geq \varepsilon^{-1}\}$
- Condition number of eigenvectors matrix $\kappa_e(A) = \|X\| \cdot \|X^{-1}\|$

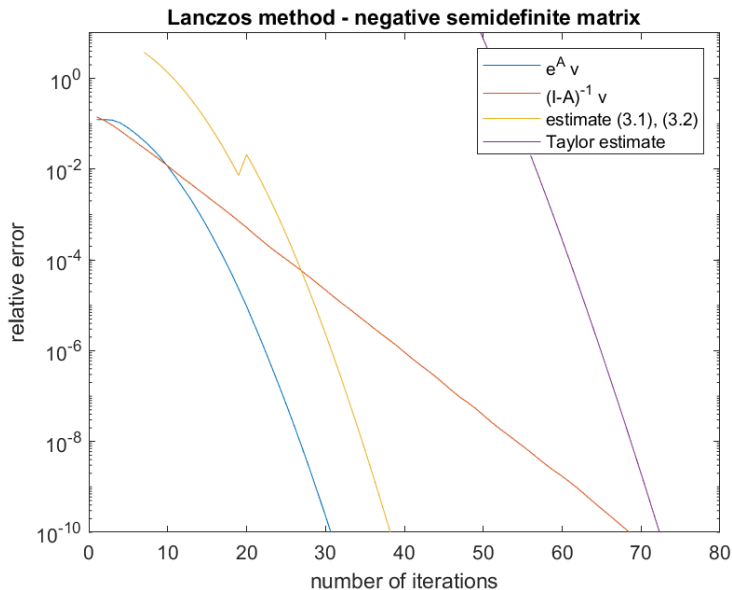
Error bounds

- $\Lambda_e(A) \subset E, \Lambda_\gamma(A) \cup \Lambda_\gamma(H_m) \subset G$
- Error bound lemma with $M = \frac{1}{2}(1 + \|V_m\| \cdot \|D_m^{-1}W_m^*\|)I(\partial E)/(\varepsilon\gamma)$
- A diagonalizable, spectrum $\Lambda(A) \subset E, \Lambda(H_m) \subset G$,
 $\|(\lambda I - H_m)^{-1}\| \leq \gamma^{-1}$
- Error bound lemma with $M = 3\kappa_\varepsilon(A)(\delta^{-1} + \|V_m\| \cdot \|D_m^{-1}W_m^*\|\gamma^{-1})$,
 δ is distance between $\Lambda(A)$ and Γ

Lanczos method - bound for skew-Hermitian matrix



Lanczos method - bound for negative semidefinite matrix



Examples

No graphs in paper, consider some examples to show bounds.

Last section about ODE schemes could be omitted, no graphs just scheme definition.